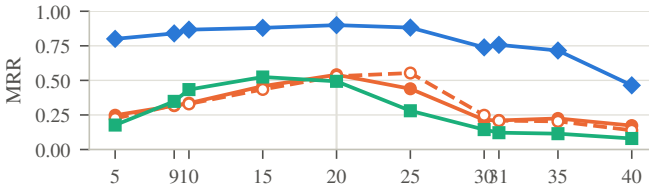
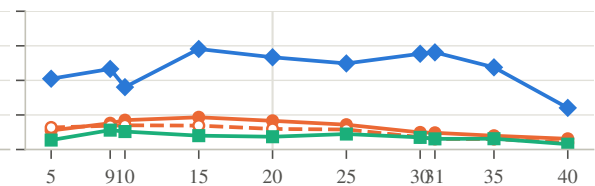
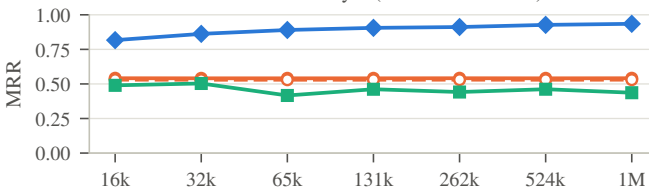
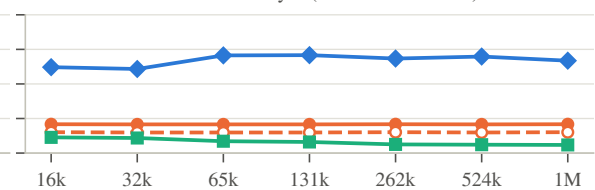


Lean 4 $\rightarrow$ HOL Light (317 problems)Lean 4 $\rightarrow$ Metamath (482 problems)

Gemma-2-9B layer (131k-feature SAE)



Gemma-2-9B layer (131k-feature SAE)



SAE width at layer 20 (features; average L0 48–58)

—●— Dense

—○- Dense, PC1 removed

—■— SAE, unweighted

—◆— SAE-IDF (ours)

SAE width at layer 20 (features; average L0 48–58)